

KRINS ITALIYA

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Education

Bachelor of Technology in Electronics and Communication Engineering (ECE) 2023 – Present

Pandit Deendayal Energy University (PDEU), Gujarat

CGPA: **9.24**

Expected Graduation: June 2027

Relevant Coursework: Digital Logic Design, Computer Design and Architecture, Circuit Analysis and PCB Design Fundamentals

Technical Skills

Programming & HDL: C Programming, Verilog HDL (RTL Modeling), 8086 Assembly Language

Digital & VLSI Design: Combinational and Sequential Logic, FSM Design, RTL Design and Verification, ASIC Design Flow (RTL-to-GDSII), Basic Timing Concepts

Embedded Systems: Microcontroller-Based Systems, Interrupt Handling, Timer Modules, Sensor and Actuator Interfacing

EDA Tools & Platforms: Proteus, MATLAB, Synopsys Design Compiler, Linux, KiCad, LTspice

Projects

RTL-to-GDSII Implementation of Synchronous 4-bit Adder–Subtractor Jun 2025

- Designed a synchronous 4-bit adder–subtractor at RTL level using Verilog HDL.
- Developed a self-checking testbench and verified functionality through RTL simulation.
- Performed logic synthesis and explored physical design stages including floorplanning, placement, and routing.
- Analyzed post-synthesis timing behavior and functional correctness using waveform analysis.

Interrupt-Driven LED Pattern Generator using 8086 and 8255 Nov 2025

- Designed an LED pattern generator using 8086 microprocessor with 8255 PPI.
- Implemented interrupt-driven control logic for efficient and delay-free operation.
- Used external latch for address and data demultiplexing.
- Simulated and verified system behavior using Proteus.

RISC-V Reference SoC Design & Tapeout Flow (VSD Workshop) Aug 2025

- Executed an end-to-end RTL-to-GDSII design flow for a RISC-V based reference SoC.
- Verified RTL functionality through simulation and waveform analysis.
- Performed logic synthesis and evaluated design quality using timing and resource metrics.
- Gained exposure to industry-oriented SoC design and tapeout preparation flow.

ESP32-Based Intelligent Robotic Control Systems Apr 2024

- Developed ESP32-based robotic systems including hand-gesture controlled and obstacle-avoiding robots.
- Interfaced sensors, motor drivers, and actuators for real-time motion control.
- Implemented embedded control logic and validated system performance through testing.

Achievements

- Completed RTL-to-GDSII ASIC Design Workshop with hands-on RTL and synthesis experience.
- Secured **1st Place** at RoboChef Workshop for ESP32-based robotic system.
- Designed and verified multiple digital RTL modules using Verilog HDL.
- Implemented interrupt-driven embedded system applications using 8086 microprocessor.

Technical Interests

VLSI System Design, ASIC Design Flow, RTL Design and Verification, Embedded Systems, Hardware–Software Co-Design, Digital System Architecture